PRACTICAL-5

AIM:

To build a simple rule-based expert system for laptop troubleshooting using IF-THEN rules.

Program:

```
# Simple Rule-Based Expert System
# Laptop Troubleshooting Expert System

def laptop_expert(turns_on, internet_works, is_slow):

    if turns_on == "no":
        return "Check power cable of the laptop"

    elif internet_works == "no":
        return "Check WiFi and restart the router"

    elif is_slow == "yes":
        return "Close unused programs and restart the system"

    else:
        return "No common problem in the expert system"

# Ask the user for facts

turns_on = input("Does the laptop turn on? yes/no: ").lower()

internet_works = input("Does the internet work? yes/no: ").lower()

is_slow = input("Is the laptop slow? yes/no: ").lower()

# Get advice from expert system

advice = laptop_expert(turns_on, internet_works, is_slow)

# Display result

print("\nExpert System Advice")
print(advice)
```

Output:

Does the laptop turn on? yes/no: yes
Does the internet work? yes/no: no
Is the laptop slow? yes/no: no

Expert System Advice